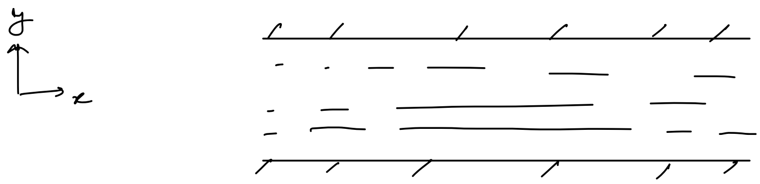
Problem 1 (two dimensional uni-directional flow).



at rest $y=H$, $\forall t$

I.C $V_x=0$ $\forall y$

$\rightarrow U_0$ at $y=0$ $\forall t$

Governing eq'n:

$$\rho\left(\frac{\partial v_x}{\partial t} + \underbrace{v_x\cancel{\frac{\partial v_x}{\partial x}}}_{0} + \underbrace{\cancel{v_y}}_{0}\frac{\partial v_x}{\partial y} + \underbrace{\cancel{v_z}}_{0}\frac{\partial v_x}{\partial z}\right) = \underbrace{-\cancel{\frac{\partial p}{\partial x}}}_{0} + \mu\left(\underbrace{\cancel{\frac{\partial^2 v_x}{\partial x^2}}}_{0} + \frac{\partial^2 v_x}{\partial y^2} + \underbrace{\cancel{\frac{\partial^2 v_x}{\partial z^2}}}_{0}\right) + \underbrace{\cancel{\rho g_x}}_{0}$$

fully developed along x-dirc'n — not pressure driven — no body force

$$\rho\frac{\partial v_x}{\partial t} = \mu\frac{\partial^2 v_x}{\partial y^2} \Rightarrow \boxed{\frac{\partial v_x}{\partial t} = \nu\frac{\partial^2 v_x}{\partial y^2}}$$

non-dimensionalisation

$$v_x^* = \frac{v_x}{U_0}, \quad y^* = \frac{y}{H}, \quad t^* = \frac{t}{T}$$

$$t^* = \frac{t}{(H^2/\nu)}$$

To find characteristic time scale we perform order of magnitude analysis.

$$\frac{\partial v_x}{\partial t} = \nu\frac{\partial^2 v_x}{\partial y^2}$$

$$\frac{U_0}{T} = \nu\frac{U_0}{H^2} \Rightarrow T = \frac{H^2}{\nu}$$

substituting variables

$$\Rightarrow \frac{\partial(v_x^* U_0)}{\partial(T t^*)} = \nu\frac{\partial^2(v_x^* U_0)}{\partial(y^* H)^2}$$

$$\Rightarrow \cancel{\frac{U_0}{T}}\frac{\partial v_x^*}{\partial t^*} = \cancel{\frac{\nu U_0}{H^2}}\frac{\partial^2 v_x^*}{\partial y^{*2}}$$

boundary & initial conditions in non-dimensionalized form.

$y=H$, $y^*=1$, $v_x^*=0$

$y=0$, $y^*=0$, $v_x^*=1$

$$\boxed{\frac{\partial v_x^*}{\partial t^*} = \frac{\partial^2 v_x^*}{\partial y^{*2}}}$$

$\rightarrow$ (NON-DIMENSIONALISED).

Let us claim that the solution is in the following form:-

$$v_x^*(y) = \underbrace{v_{ss}(y)}_{\text{steady state solution}} + v_t(y)$$

solving at steady state $\Rightarrow$

$$\underbrace{\cancel{\frac{\partial v_{ss}}{\partial t^*}}}_{0} = \frac{\partial^2 v_{ss}}{\partial y^{*2}} \Rightarrow v_{ss} = a y^* + b$$

$1 = b$, $a + b = 0$ $\Rightarrow$ $V_{ss} = 1 - y^*$

Now, we have $V_x^* = (1 - y^*) + V_t$

$$\frac{\partial V_x^*}{\partial t^*} = \frac{\partial^2 V_x^*}{\partial y^{*2}} \Rightarrow \frac{\partial (1 - y^* + V_t)}{\partial t^*} = \frac{\partial^2 (1 - y^* + V_t)}{\partial y^{*2}}$$

$$\Rightarrow \frac{\partial V_t}{\partial t^*} = \frac{\partial^2 V_t}{\partial y^{*2}}$$

B.C

$1 - y^* + V_t = 1$ at $y^* = 0$ $\Rightarrow$ $V_t = 0$

$1 - y^* + V_t = 0$ at $y^* = 1$ $\Rightarrow$ $V_t = 0$

I.C

$1 - y^* + V_t = 0$ $\forall y^*$

$V_t = y^* - 1$ (not physically realizable)

Solving for V_t using separation of variable:-

$V_t(y^*) = f(y^*)\, g(t^*)$ [separation of variable]

$$f g' = g f''$$

$$\frac{g'}{g} = \frac{f''}{f} = -c^2$$

↳ negative constant

$\frac{g'}{g} = (-c^2)$

$$\boxed{g = A e^{-c^2 t}}$$

$\frac{f''}{f} = -c^2 \Rightarrow f'' + c^2 f = 0$

$$\frac{d^2 f}{\partial y^2} + c^2 f = 0$$

$$\boxed{f(y^*) = B\sin(cy^*) + D\cos(cy^*)}$$

$\Rightarrow$ $V_t = 0$ for both $y^* = 0$, $y^* = 1$

$f(y^*)\, g(t^*) = 0$

$g(t^*)$ ↓ function of time (cannot be zero always)

hence $f(y^*) = 0$ $\forall t$

applying B.C.

$0 = 0 + D$ $\Rightarrow D = 0$

$0 = B \sin c$ $\Rightarrow \sin c = 0$ $\Rightarrow c = n\pi$

$n = -\infty$ to $+\infty$

$$f(y^*) = \sum_{n=-\infty}^{+\infty} B_n \sin(n\pi y^*)$$

$$v_t(y^*, t^*) = \sum A_n e^{-Cn^2 t^*} \cdot B_n \sin(n\pi y^*)$$

$$= \sum_{-\infty}^{+\infty} D_n \cdot e^{-Cn^2 t^*} \cdot \sin(n\pi y^*)$$

$$V_t(y^*, t^*) = \sum_{n=1}^{+\infty} Q_n \cdot e^{-Cn^2 t^*} \cdot \sin(n\pi y^*)$$

$Q_n = D_n + D_{-n}$
$n = 0$ (does not contribute)

Applying $\underline{\underline{I.C}}$

$$y^* - 1 = \sum_{n=1}^{\infty} Q_n \cdot \sin(n\pi y^*)$$

$$\int_0^1 (y^* - 1) \sin(m\pi y^*) dy = \int_0^1 \sum_{n=1}^{\infty} Q_n \cdot \sin(n\pi y^*) \cdot \sin(m\pi y^*) \, dy$$

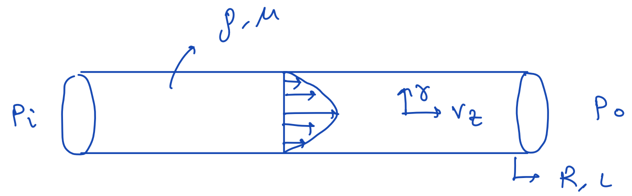
$$\int_0^1 (y^* - 1) \sin(m\pi y^*) \, dy = \sum_{n=1}^{\infty} Q_n \int_0^1 \sin(n\pi y^*) \cdot \sin(m\pi y^*) \, dy$$

only $m = n$ term left

$$\boxed{Q_n = \frac{-2}{n\pi}}$$

$$\therefore V(y,t) = \sum_{n=1}^{\infty} \left[\underbrace{\left(-\frac{2}{n\pi} \right) e^{(-n^2\pi^2 t / (H^2/\nu))} \sin\left(n\pi \frac{y}{H}\right)}_{\text{Unsteady}} \right] + \underbrace{\left(1 - \frac{y}{H}\right)}_{\text{Steady}}$$

Problem 2 (Start-up of laminar Flow)



① pressure driven unidirectional flow
② $\partial v_z / \partial z = 0$ (fully developed in axial direction)

We obtain the governing equation:

$$\rho \frac{\partial v_z}{\partial t} = -\frac{\partial p}{\partial z} + \mu \frac{1}{r}\left(\frac{\partial}{\partial r}\left(r \frac{\partial v_z}{\partial r}\right)\right)$$

$$\frac{\partial p}{\partial z} = \frac{P_o - P_i}{L}$$

(P is a decreasing function of z).

How is this pressure gradient linear ??

I.C $\quad v_z(t=0, r) = 0$

B.C $\quad v_z(r=R, \forall t) = 0$ (no slip)

$v_z(r=0, \forall t) = \text{finite}$

$\left.\frac{\partial v_z}{\partial r}\right|_{r=0} = 0$ (from symmetry)

Order of magnitude analysis ⇒

$$\underbrace{\rho \frac{v_z^*}{t^*} \sim \frac{P_i - P_o}{L} \sim \frac{\mu}{R}\left(\frac{1}{R}\left(\frac{R v_z^*}{R}\right)\right)}_{\text{All must have same order of magnitude.}}$$

① ② ③

Considering ① & ③ term

$$\rho \frac{\cancel{v_z^*}}{t^*} \sim \frac{\mu}{R^2} \cancel{v_z^*} \Rightarrow t^* \simeq R^2 / (\mu/\rho)$$

Considering ② & ③ term

$$\frac{P_i - P_o}{L} \sim \frac{\mu}{R^2} v_z^* \Rightarrow v_z^* \simeq \frac{(P_i - P_o) R^2}{\mu L}$$

We have now found the terms (non-dimensional) parameters for t & V_z as well.

Thus,

$$\eta = \frac{r}{R}, \quad \tau = \frac{t}{t^*}, \quad \phi = \frac{V_z}{V_z^*}$$

$$\rho \frac{\partial V_z}{\partial t} = -\frac{\partial p}{\partial z} + \frac{\mu}{r}\left(\frac{\partial}{\partial r}\left(r \frac{\partial V_z}{\partial r}\right)\right)$$

$$\rho \frac{\partial(\phi V_z^*)}{\partial(\tau t^*)} = -\left(\frac{P_o - P_i}{L}\right) + \frac{\mu}{\eta R}\left(\frac{\partial}{\partial(\eta \cancel{R})}\left(\eta \cancel{R} \frac{\partial(\phi V_z^*)}{\partial(\eta R)}\right)\right)$$

$$\frac{V_z^*}{t^*} \rho \frac{\partial \phi}{\partial \tau} = \frac{P_i - P_o}{L} + \frac{\mu V_z^*}{\eta R^2} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

$$\frac{\rho}{t^*} \frac{\partial \phi}{\partial \tau} = \frac{P_i - P_o}{V_z^* L} + \frac{\mu}{\eta R^2} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

We know $t^* = \frac{R^2 \rho}{\mu}$, $V_z^* = \frac{(P_i - P_o) R^2}{\mu L}$

we will consider

$$V_z^* = \frac{(P_i - P_o) R^2}{4 \mu L}$$

$$\frac{\cancel{\rho} \mu}{R^2 \cancel{\rho}} \frac{\partial \phi}{\partial \tau} = \frac{4\mu}{R^2} + \frac{\mu}{R^2} \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

$$\boxed{\frac{\partial \phi}{\partial \tau} = 4 + \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)}$$

Now, we assume $\phi = \phi_s + \phi_t$

Solving for steady state solution:-

$$\cancelto{0}{\frac{\partial \phi_{ss}}{\partial \tau}} = 4 + \frac{1}{\eta} \frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

$$\frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right) = -4\eta$$

$$\eta \frac{\partial \phi}{\partial \eta} = -\frac{4\eta^2}{2} + c$$

$$\frac{\partial \phi}{\partial \eta} = -2\eta + \frac{c}{\eta} \Rightarrow \phi = -\eta^2 + c \ln \eta + c_1$$

BC: As $r \to 0$, V_z is finite $\Rightarrow \therefore c = 0$

As $\eta = 1$, $\phi = 0 \Rightarrow \therefore c_1 = 1$

$$\boxed{\phi_{ss} = 1-\eta^2} \quad \text{steady state solution.}$$

→ Now, we will put $\phi_{ss}+\phi_t$ in governing equation

$$\frac{\partial \phi}{\partial \tau} = 4 + \frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta \frac{\partial \phi}{\partial \eta}\right)$$

$$\frac{\partial}{\partial \tau}(\phi_{ss}+\phi_t) = 4 + \frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta \frac{\partial}{\partial \eta}(\phi_{ss}+\phi_t)\right)$$

$$\frac{\partial \phi_{ss}}{\partial \tau} + \frac{\partial \phi_t}{\partial \tau} = 4 + \frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta\left(\frac{\partial \phi_{ss}}{\partial \eta} + \frac{\partial \phi_t}{\partial \eta}\right)\right)$$

$$\frac{\partial \phi_t}{\partial \tau} = 4 + \frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta\left(-2\eta + \frac{\partial \phi_t}{\partial \eta}\right)\right)$$

$$\frac{\partial \phi_t}{\partial \tau} = 4 + \frac{1}{\eta}\frac{\partial}{\partial \eta}\left[-2\eta^2 + \eta\frac{\partial \phi_t}{\partial \eta}\right]$$

$$\frac{\partial \phi_t}{\partial \tau} = 4 + \frac{1}{\eta}\left(-4\eta + \frac{\partial}{\partial \eta}\left(\eta\frac{\partial \phi_t}{\partial \eta}\right)\right)$$

$$\frac{\partial \phi_t}{\partial \tau} = \frac{1}{\eta}\frac{\partial}{\partial \eta}\left(\eta\frac{\partial \phi_t}{\partial \eta}\right)$$

I.C: $\phi(\eta, \tau=0) = 0$

$\phi_{ss} + \phi_t(\eta, \tau=0) = 0$

$\phi_t(\eta, \tau=0) = \eta^2 - 1$

Also, overall B.C & steady state B.C. remain equal

Hence,

$\phi_t(\eta=0, \forall \tau) = 0$

$\phi_t(\eta=0, \forall \tau) = \text{finite}$

Now applying separation of variable :-

$$\phi_t = f(\eta)g(\tau)$$

$$f g' = \frac{1}{\eta}\frac{\partial}{\partial \eta}(\eta g f')$$

$$\frac{g'}{g} = \frac{1}{\eta f}\frac{\partial}{\partial \eta}(\eta f') = -c^2$$

$$\underline{\underline{g = De^{-c^2\tau}}}$$

$$\frac{1}{\eta f}\frac{\partial}{\partial \eta}(\eta f') = -c^2$$

$$\frac{1}{\eta f} f' + \frac{1}{\eta f}\eta f'' = -c^2$$

$$\Rightarrow \frac{1}{\eta}\frac{\partial f}{\partial \eta} + \frac{1}{\cancel{\eta}}\cancel{\eta}\frac{\partial^2 f}{\partial \eta^2} + c^2 f = 0$$

$$\Rightarrow \eta^2 \frac{\partial^2 f}{\partial \eta^2} + \eta \frac{\partial f}{\partial \eta} + c^2\eta^2 f = 0$$

$$\Rightarrow (\eta c)^2 \frac{\partial^2 f}{\partial (\eta c)^2} + (\eta c)\frac{\partial f}{\partial (\eta c)} + (c^2\eta^2) f = 0$$

$$\eta c = q$$

$$q^2 \frac{\partial^2 f}{\partial q^2} + q\frac{\partial f}{\partial q} + (q^2 - 0^2) f = 0$$

Comparing it with standard Bessel Differential Equation we get 0th order solution.